NGUYEN QUOC HOAN

RESEARCH TRAINEE

OBJECTIVE

Develop a professional research mindset, strengthen self-learning abilities, and build a solid foundation for tackling real-world problems in Automated Software Engineering and Automated AI Engineering.

CORE COMPETENCIES

- *Programming & Algorithmic Thinking*
- *Languages: Python (Basic), C/C++, Java*
- *English: Technical Reading & Research Paper Comprehension*
- *Self-study & Independent Research Skills*
- *Tools: Git (basic), NumPy, Matplotlib (Beginner), VS Code*

CONTACT:

Phone: 0368 665 348
Email: 24020135@vnu.edu.vn
nguyenquochoan2006@gmail.com

EDUCATION

University of Engineering and Technology – Vietnam National University, Hanoi

MAJOR: INFORMATION TECHNOLOGY
CUMULATIVE GPA: 3.03/4.0
RELEVANT COURSEWORK:

- CALCULUS 2: A
- LINEAR ALGEBRA: A
- ADVANCED PROGRAMMING: B+

ACADEMIC PROJECTS

1. **BallWarrior - Arkanoid Style Game (Java/Swing)**
   - Description: Developed a highly extensible brick-breaking game using OOP principles and Design Patterns (State, Strategy, Factory).
   - Key Features:
     - Architecture: Decoupled game logic from UI using a modular structure (Core, Entity, UI); implemented custom Vector2D physics and Collision System (Circle-AABB resolution).
     - Advanced Systems: Built a Replay System (recording input frames/seeds), Skill System (Polymorphic Active/Passive skills), and Gacha mechanics.
     - Tooling: Created a GUI-based Level Editor supporting JSON serialization for map saving/loading.
   - Role: Team member.

2. **Dragon Ball: Fireball Fury (C++/SDL)**
   - Description: Built a 2D platformer engine from scratch focusing on performance and architecture without using commercial game engines.
   - Key Technical Highlights:
     - Custom ECS: Implemented Entity-Component-System architecture to manage game objects efficiently.
     - Core Systems: Developed Finite State Machine (FSM) for character states, manual collision detection algorithms, and resource management.
   - Role: Solo developer.

SELF-STUDY & RESEARCH INTERESTS

- Machine Learning Fundamentals: Currently self-studying basic concepts (Linear Regression, Gradient Descent).
- Technology Exploration: Regularly reading technical documentation for Python libraries (NumPy, Pandas) to deepen understanding.